Question 1:

Find $\int \frac{1}{x^2-16}dx$

Hint: $\int \frac{1}{x}dx = \ln|x| + C;\quad \int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|ax \pm b| + C;$

a) Decompose $\dfrac{1}{x^2-16} = \dfrac{?}{x+4} + \dfrac{?}{x-4}$

b) Answer: $\int \frac{1}{x^2-16}dx =$ ______________

Question 2:

Find $\int \frac{15}{x^2+3x+2}dx$

Hint: $\int \frac{1}{x}dx = \ln|x| + C$; $\int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|ax \pm b| + C$;

a) Decompose $\frac{15}{x^2+3x+2} = \frac{?}{x+2} + \frac{?}{x+1}$

b) Answer: $\int \frac{15}{x^2+3x+2}dx =$ ________________

Question 3:

Find $\int \frac{x^2+3x+4}{x^3-4x}dx$

Hint: $\int \frac{1}{x}dx = \ln|x| + C$; $\quad \int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|ax \pm b| + C$;

a) Decompose $\frac{x^2+3x+4}{x^3-4x} = \frac{?}{x} + \frac{?}{x+2} + \frac{?}{x-2}$

b) Answer: $\int \frac{x^2+3x+4}{x^3-4x}dx =$ ______________

Question 4:

Find $\int \frac{x^3 - 3x^2 - 15x + 5}{x - 8}\, dx$

Hint: $\int \frac{1}{x} dx = \ln|x| + C;$ $\quad \int \frac{1}{ax \pm b} dx = \frac{1}{a} \ln|ax \pm b| + C;$

a) Divide $\frac{x^3 - 3x^2 - 15x + 5}{x - 8} = ?$

b) $\int \frac{x^3 - 3x^2 - 15x + 5}{x - 8}\, dx$ = ____________________

Question 5:

Find $\int \frac{5x^2+7x-1}{x^3+x^2}dx$

Hint: $\int \frac{1}{x}dx = \ln|x| + C$; $\int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|ax \pm b| + C$;

a) Decompose $\frac{5x^2+7x-1}{x^3+x^2} = \frac{5x^2+7x-1}{x^2(x+1)} = \frac{Ax+B}{x^2} + \frac{C}{x+1} = \frac{?}{x^2} + \frac{?}{x+1}$

b) Answer: $\int \frac{5x^2+7x-1}{x^3+x^2}dx =$ ____________________

Question 6:

Find $\int \frac{x^2+5x-2}{x^3-4x^2+4x}dx$

Hint: $\int \frac{1}{x}dx = \ln|x| + C$; $\int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|ax \pm b| + C$;

a) Decompose $\frac{x^2+5x-2}{x^3-4x^2+4x} = ?$

Note: $\frac{x^2+5x-2}{x^3-4x^2+4x} = \frac{x^2+5x-2}{x\left(x^2-4x+4\right)} = \frac{x^2+5x-2}{x(x-2)(x-2)} = \frac{x^2+5x-2}{x(x-2)^2}$

$(x-2)^2$ is considered a repeated linear factor.

Rule for repeated linear factor is as follows:

$$\frac{x^2+5x-2}{x(x-2)^2} = \frac{A}{x} + \frac{B}{(x-2)} + \frac{C}{(x-2)^2} = \frac{A(x-2)^2 + B \cdot x \cdot (x-2) + C \cdot x}{x(x-2)^2}$$

Now find A, B, and C.

b) Answer: $\int \frac{x^2+5x-2}{x^3-4x^2+4x}dx =$ ______________

Question 7:

Find $\int \frac{x^2 - 9}{x^3 + x} dx$

Hint: $\int \frac{1}{x} dx = \ln|x| + C; \quad \int \frac{1}{ax \pm b} dx = \frac{1}{a} \ln|ax \pm b| + C;$

a) Decompose $\frac{x^2 - 9}{x^3 + x} = \frac{x^2 - 9}{x^3 + x^2} = \frac{x^2 - 9}{x^2(x+1)} = \frac{Ax + B}{x^2} + \frac{C}{x+1} = \frac{?}{x^2} + \frac{?}{x+1}$

b) Answer: $\int \frac{x^2 - 9}{x^3 + x} dx =$ ______________________________